C1.2 Atomic structure

Summary
An atom is the smallest component of an element that gives an element its property. These properties can be explained by models of atomic structure. Current models suggest that atoms are made of smaller sub-atomic particles called protons, neutrons and electrons. They suggest that atoms are composed of a nucleus surrounded by electrons. The nucleus is composed of neutrons and protons. Atoms of each element have the same number of protons as electrons. Atoms of different elements have different numbers of protons. Atoms of the same element will have the same number of protons but may have different numbers of neutrons.

Underlying knowledge and understanding
Learners should be familiar with the simple (Dalton) atomic model.

Common misconceptions
Learners commonly have difficulty understanding the concept of isotopes due to the fact they think that neutral atoms have the same number of protons and neutrons. They also find it difficult to distinguish between the properties of atoms and molecules. Another common misconception is that a positive ion gain protons or a negative ion loses protons i.e. that there is a change in the nucleus the atom rather than a change in the number of electrons.

Tiering
Statements shown in **bold** type will only be tested in the Higher Tier papers.
All other statements will be assessed in both Foundation and Higher Tier papers.

Reference	Mathematical learning outcomes	Mathematical skills
CM1.2i	relate size and scale of atoms to objects in the physical world	M4a
CM1.2ii ☑	estimate size and scale of atoms and nanoparticles	M1c

Topic content			Opportunities to cover:		Practical suggestions
Learning outcomes		To include	Maths	Working scientifically	
C1.2a	describe how and why the atomic model has changed over time	the models of Dalton, Thomson, Rutherford, Bohr, Geiger and Marsden		WS1.1a, WS1.1i, WS1.2b	Timeline of the atomic model.
C1.2b	describe the atom as a positively charged nucleus surrounded by negatively charged electrons, with the nuclear radius much smaller than that of the atom and with most of the mass in the nucleus			WS1.4a	
C1.2c	recall the typical size (order of magnitude) of atoms and small molecules	the concept that typical atomic radii and bond length are in the order of 10^{-10}m	M1c, M4a	WS1.1c, WS1.4b, WS1.4c, WS1.4d, WS1.4e, WS1.4f	
C1.2d	recall relative charges and approximate relative masses of protons, neutrons and electrons			WS1.4a, WS1.4b, WS1.4c	
C1.2e	calculate numbers of protons, neutrons and electrons in atoms and ions, given atomic number and mass number of isotopes	definitions of an ion, atomic number, mass number and an isotope, also the standard notation to represent these		WS1.3c, WS1.4b	